Limites e L'Hôpital

Limites fundamentais

$\lim_{x\to 0} \frac{\sin x}{x} = 1$

$\lim_{x\to 0} \frac{e^x - 1}{x} = 1$

$\lim_{x\to 0} \frac{\ln(1+x)}{x} = 1$

$\lim_{x\to 0} (1+x)^{1/x} = e$

$\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}$

$\lim_{x\to \infty} \left(1 + \frac{a}{x}\right)^x = e^a$

Regra de L'Hôpital

Se $\frac{0}{0}$ ou $\frac{\infty}{\infty}$, então $\lim \frac{f}{g} = \lim \frac{f'}{g'}$ (se existir).

Derivadas

Regras básicas

$\frac{d}{dx} c = 0$

$\frac{d}{dx}[cf(x)] = cf'(x)$

$\frac{d}{dx}[f \pm g] = f' \pm g'$

$\frac{d}{dx}[fg] = f'g + fg'$

$\frac{d}{dx}\left[\frac{f}{g}\right] = \frac{f'g - fg'}{g^2}$

Potência/Exp/Log

$\frac{d}{dx} x^n = nx^{n-1}$

$\frac{d}{dx} e^x = e^x$

$\frac{d}{dx} a^x = a^x \ln a$

$\frac{d}{dx} \ln x = \frac{1}{x}$

Trigonométricas

$\frac{d}{dx} \sin x = \cos x$

$\frac{d}{dx} \cos x = -\sin x$

$\frac{d}{dx} \tan x = \sec^2 x$

$\frac{d}{dx} \cot x = -\csc^2 x$

Inversas e Hiperbólicas

$\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}}$

$\frac{d}{dx} \arccos x = -\frac{1}{\sqrt{1-x^2}}$

$\frac{d}{dx} \arctan x = \frac{1}{1+x^2}$

$\frac{d}{dx} \sinh x = \cosh x$

$\frac{d}{dx} \cosh x = \sinh x$

Integrais

Básicas

$\int k\,dx = kx + C$

$\int x^n dx = \frac{x^{n+1}}{n+1} + C \ (n \neq -1)$

$\int \frac{1}{x} dx = \ln|x| + C$

Exponenciais

$\int e^x dx = e^x + C$

$\int a^x dx = \frac{a^x}{\ln a} + C$

Trigonométricas

$\int \sin x dx = -\cos x + C$

$\int \cos x dx = \sin x + C$

$\int \tan x dx = \ln|\sec x| + C$

$\int \sec^2 x dx = \tan x + C$

$\int \csc^2 x dx = -\cot x + C$

$\int \sec x \tan x dx = \sec x + C$

Racionais e especiais

$\int \frac{1}{a^2+x^2} dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C$

$\int \frac{1}{\sqrt{a^2-x^2}} dx = \arcsin\left(\frac{x}{a}\right) + C$

$\int \frac{1}{x^2-a^2} dx = \frac{1}{2a} \ln\left|\frac{x-a}{x+a}\right| + C$

$\int \sqrt{a^2-x^2} dx =$

$\frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2} \arcsin\left(\frac{x}{a}\right) + C$

Matemática Básica

Fatoração

$(a \pm b)^2 = a^2 \pm 2ab + b^2$

$a^2 - b^2 = (a+b)(a-b)$

$a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$

Logaritmos

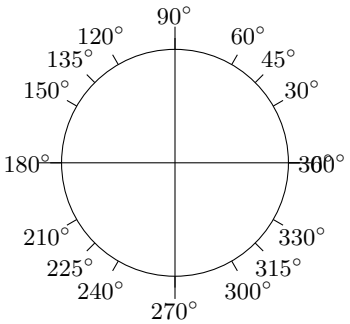
$\ln(ab) = \ln a + \ln b$

$\ln(a^b) = b \ln a$

$\ln\left(\frac{a}{b}\right) = \ln a - \ln b$

$e^{\ln a} = a$

Círculo Trigonométrico



Ângulo	sin	cos	tan
0°	0	1	0
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
45°	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90°	1	0	–
120°	$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$-\sqrt{3}$
180°	0	-1	0

Trigonometria

Fundamentais

$\sin^2 x + \cos^2 x = 1$

$1 + \tan^2 x = \sec^2 x$

Recíprocas

$\frac{1}{\cos x} = \sec x$

$\frac{1}{\sin x} = \csc x$

Soma/Subtração

$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$

$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$

Duplicação

$\sin(2x) = 2 \sin x \cos x$

$\cos(2x) = \cos^2 x - \sin^2 x$

$\cos(2x) = 2\cos^2 x - 1$

$\cos(2x) = 1 - 2\sin^2 x$

Constantes Úteis

$\pi = 3{,}14159\ldots$

$e = 2{,}71828\ldots$